
Netflix Catalogue Analysis

Data Visualisation Case Study

Exploratory analysis of a public Netflix catalogue dataset to identify catalogue growth patterns, content mix, seasonality and metadata quality issues.

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Python | Pandas | Data visualisation | Analytical storytelling

Portfolio project | May 2024

DATA

Note: this is an exploratory portfolio project using public catalogue metadata, not a causal analysis of Netflix business performance.

Project scope

A concise portfolio version of the original analysis

Objective

Use catalogue metadata to identify descriptive trends and turn them into clear visual insights for a business audience.

Dataset

Public Netflix titles dataset with metadata such as type, date added, country, rating, duration and category tags.

Tools

Python, Pandas, Matplotlib, Seaborn, Plotly, GeoPandas and Jupyter Notebook.

APPROACH

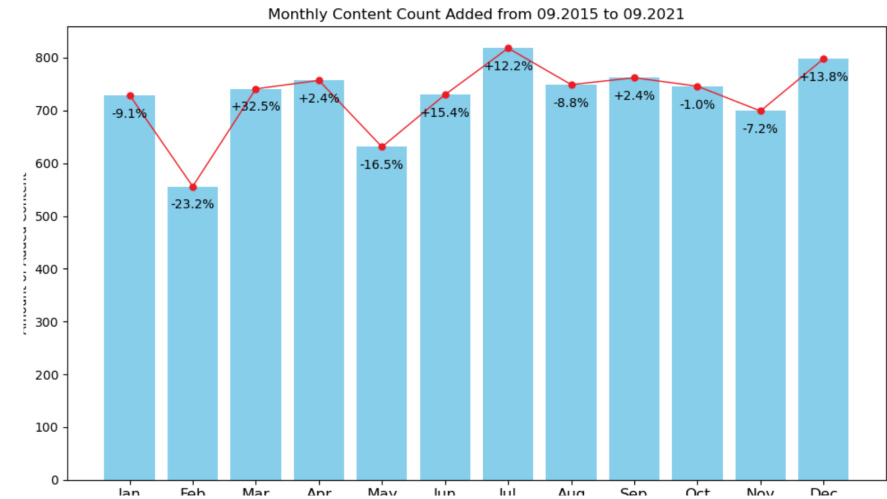
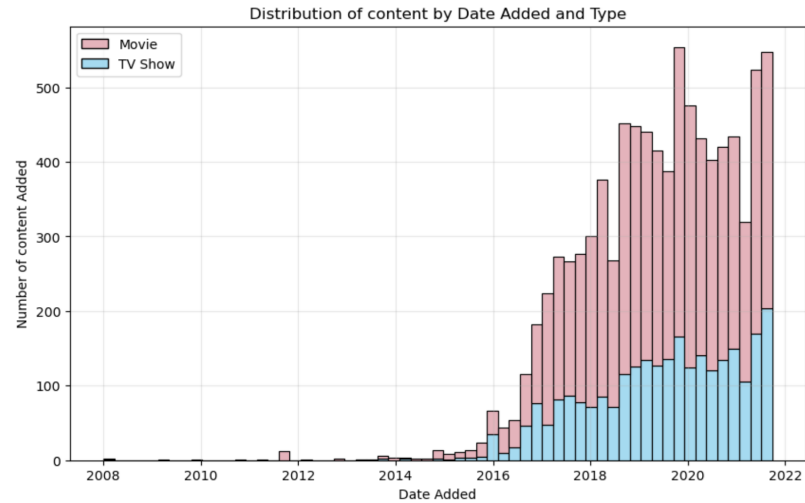
- Reviewed dataset structure, missing values, duplicates and inconsistent category fields.
- Cleaned date, country, rating and duration fields to prepare the data for analysis.
- Created visual outputs for growth over time, seasonality, content type, rating mix, genre distribution and geographic coverage.
- Converted technical findings into business-facing observations and avoided unsupported causal claims.

PORTFOLIO RELEVANCE

This project demonstrates analytical judgement, data cleaning, visual storytelling and the ability to explain data patterns to non-technical stakeholders.

Catalogue growth and seasonality

Descriptive trends from the catalogue metadata

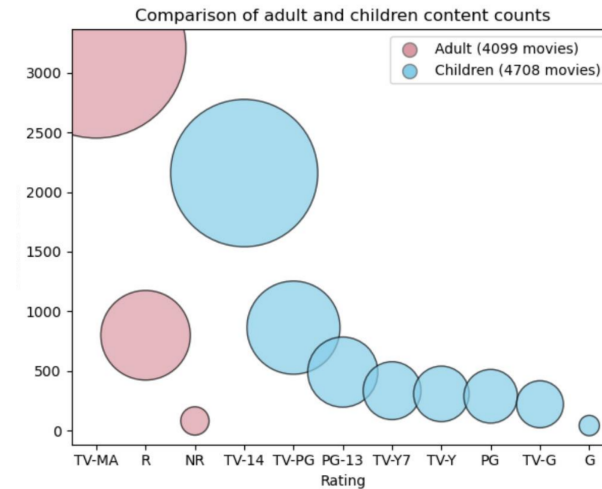
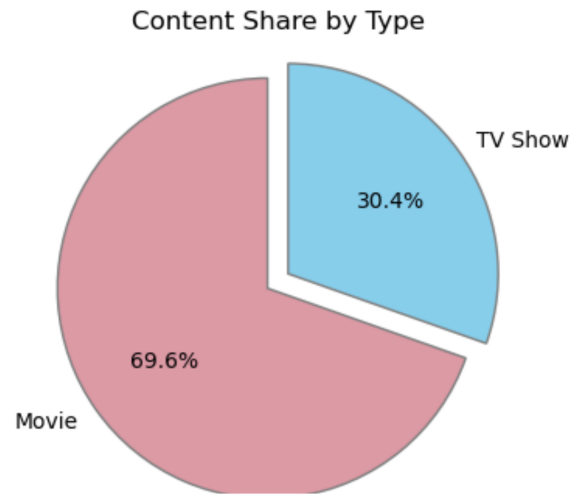


Key observations

- Catalogue additions accelerate strongly after 2015 in the dataset.
- Movies dominate additions, while TV shows form a material share of new catalogue items.
- The monthly view suggests recurring lower activity in selected months; this should be treated as a descriptive pattern requiring further validation.
- The visuals show how time-based data can be turned into a management-friendly trend view.

Content mix and audience classification

Content type and rating distributions

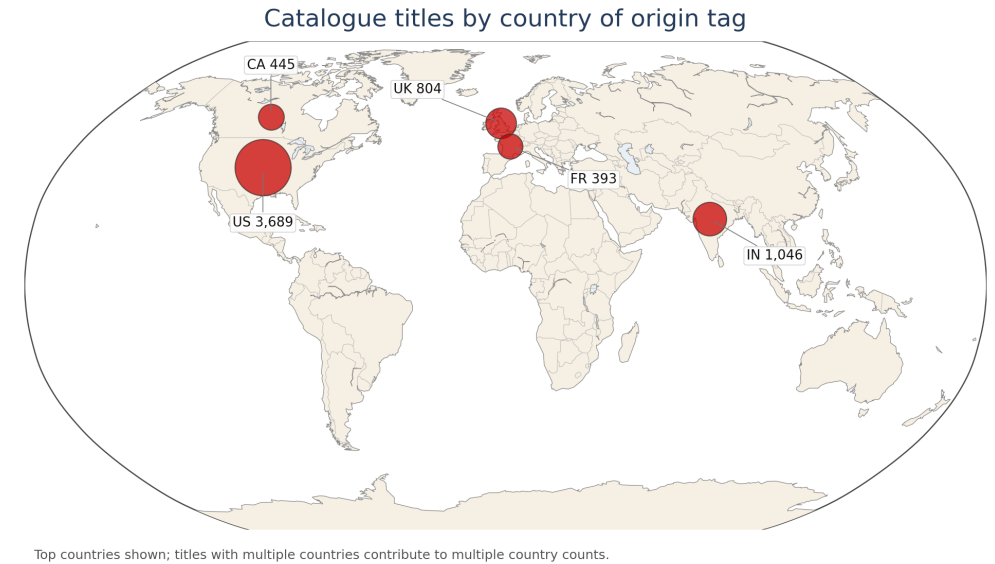
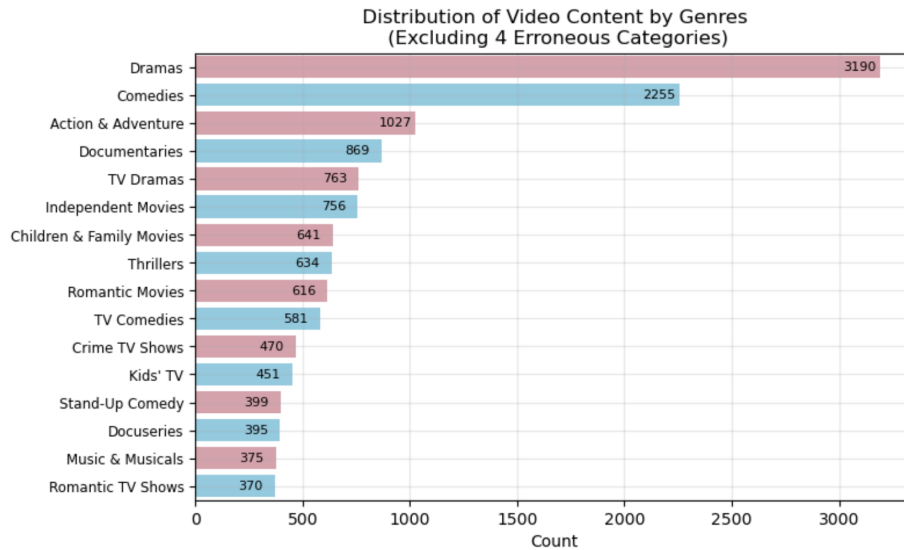


Key observations

- Movies account for approximately 69.6% of the catalogue records, while TV shows account for approximately 30.4%.
- TV-MA and TV-14 are the largest rating groups in the dataset.
- The same data can support self-serve reporting by content type, rating group, suitability category and business segment.
- The analysis highlights the importance of clear category definitions before building dashboards.

Genre and geographic coverage

Category cleaning and dimensional analysis



Key observations

- After removing metadata-like category tags, drama and comedy appear as the largest genre groups.
- The country analysis uses exploded country tags, so co-produced titles can contribute to multiple country counts.
- Using exact country tags avoids the earlier overcounting risk from substring matching.
- This part demonstrates dimensional analysis, metadata validation and geographic segmentation.

Note: Country counts use split/explode logic on the country field; records with multiple countries are counted once per listed country tag.

Data quality and analytical judgement

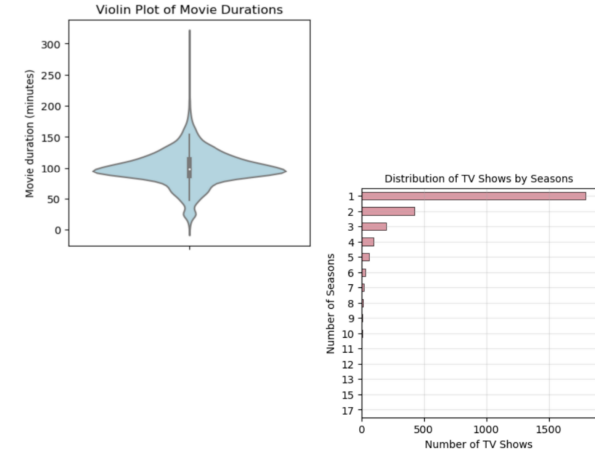
Where the work moves beyond charts

DATA PREPARATION EXAMPLES

- Converted date fields and extracted month/year features for time-based analysis.
- Filled or labelled missing values in rating, country, director and cast fields where appropriate.
- Separated movie duration in minutes from TV show duration in seasons to avoid mixing different units.
- Flagged mixed semantic tags in the genre/category field before drawing conclusions.

ANALYTICAL CONTROLS

- Kept findings descriptive and avoided claiming that catalogue patterns explain business success.
- Used visual checks to detect anomalies and inconsistencies before presenting insights.
- Focused on outputs that could be reused in self-serve reporting or dashboard requirements.



What this case study demonstrates

Skills relevant to data visualisation and automation roles

- Data cleaning and preprocessing with Python and Pandas.
- Exploratory analysis of time-series, categorical and geographic data.
- Visual storytelling with charts, captions and business-facing observations.
- Ability to turn messy metadata into structured analytical views.
- Commercial lens: focusing on usefulness, clarity and decision support rather than decorative charts.

Relevant outputs

Trend views, category breakdowns, data quality notes, dimensional summaries and portfolio-ready presentation of analysis.

Next improvement

Convert the analysis into an interactive dashboard or self-serve BI prototype with filters for date, type, rating, genre and country.

Contact

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